

cfRNA-BrainTrace Supplementary File

Single supplementary file with methods, results, figures and tables

Updated for Bioinformatics Application Note submission draft, 2026-06-19

Supplementary Methods and Results

S1 Formal validation route

All validation analyses followed the same formal route used by the software. A query profile was first represented in logCPM- or logTPM-compatible expression space and then projected into Bo2023-like VSD space for Network scoring only. The top three Networks formed the candidate beam. Resolution-group and exact-region candidates were then reranked within that beam using local logCPM-compatible expression. This design separates broad candidate generation from fine regional interpretation.

S2 Internal validation design

Internal validation used the Bo2023 macaque brain reference. Two settings were evaluated. Leave-one-sample-out validation tested whether the route could recover the held-out sample label when other samples from the reference remained available. Leave-one-monkey-out validation held out all samples from one animal at a time and therefore tested donor-level generalization. Metrics were reported separately for Network, resolution-group and exact-region levels. Top1, Top3 and median true-rank were used to distinguish exact calls from candidate-list recovery.

S3 External validation design

External validation was limited by the label resolution available in each dataset. AHBA human brain RNA-seq was used for mapped-label validation because its anatomical labels could be harmonized to Network, resolution-group and a subset of exact-region labels. TCGA/BraTS glioma tissue RNA-seq with MRI-derived labels was used only for coarse anatomical consistency because its truth labels are human imaging labels, not Bo2023 macaque exact-region identifiers. GSE189919 was used to test whether an external matrix could be projected into the model gene space; it was not used for accuracy estimation because patient-level anatomical truth was unavailable.

S4 Figures and tables

Supplementary Figure S1 provides the main route and validation summary artwork. Supplementary Tables S1-S5 provide the internal validation design, internal validation results, external validation design, external validation results and figure/table index. Together, these materials document both how the route was tested and what result each validation setting supports.

S1 Internal route selection

The first internal analysis evaluated Network-level candidate generation. Projected VSD achieved Network Top1/Top3 of 58.00%/91.58% in LOSO and 53.72%/91.33% in LOMO, exceeding logCPM baseline and native VSD at Network Top3. Direct exact-region scoring was lower and less stable, especially in LOMO. These results motivated the use of projected VSD for broad Network beam generation and logCPM-compatible expression for downstream local reranking.

S2 Formal internal three-tier validation

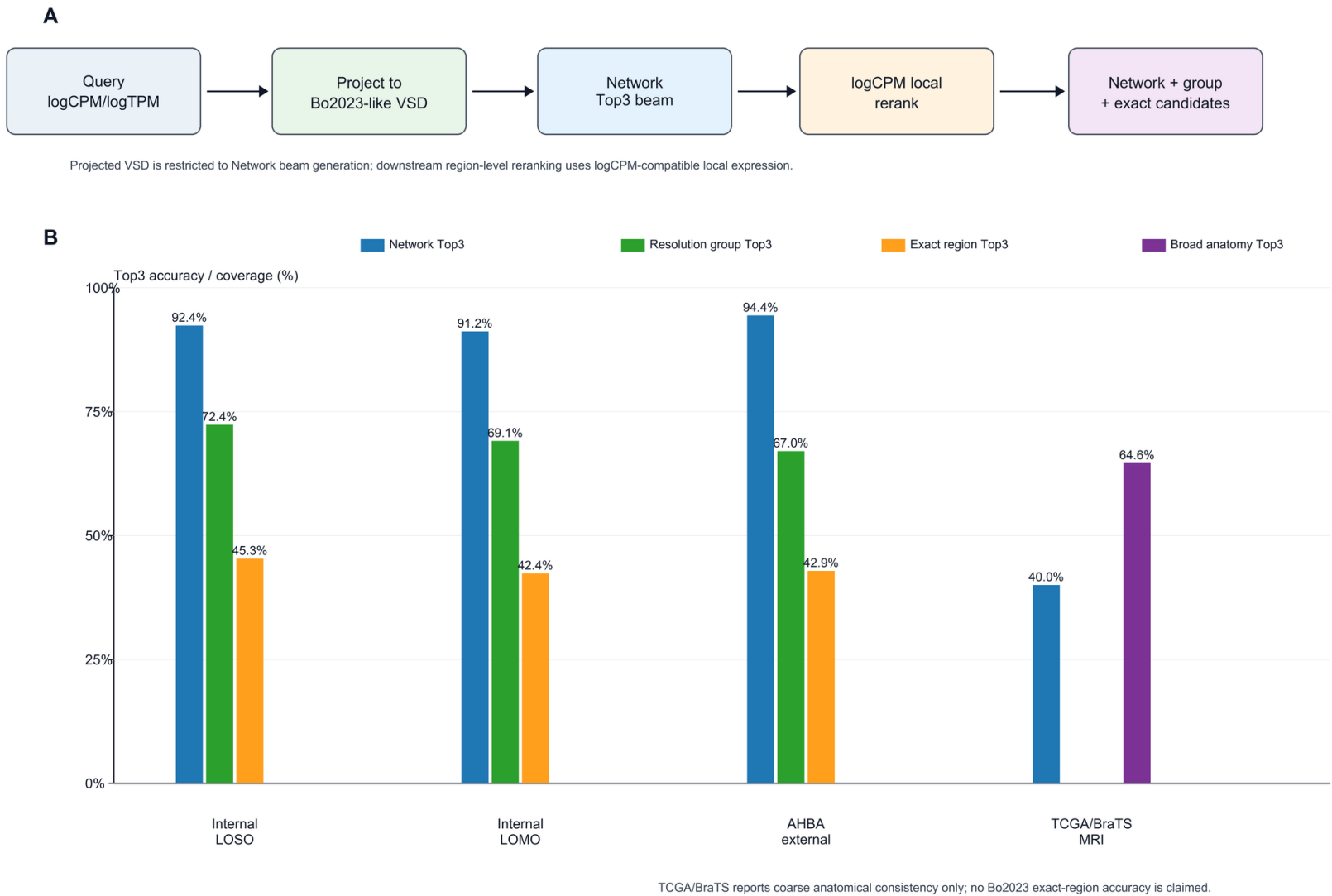
In the complete LOSO validation, the formal route achieved Network, resolution-group and exact-region Top3 values of 92.38%, 72.36% and 45.33%. In complete LOMO validation, the corresponding Top3 values were 91.21%, 69.09% and 42.36%. Median true-rank increased from Network to exact-region levels, consistent with decreasing anatomical certainty at finer resolution. Resolution group is therefore the preferred region-level endpoint, while exact-region output is retained as a candidate ranking.

S3 External validation results

In AHBA, the hybrid route achieved Network Top1/Top3 of 74.68%/94.42%, resolution-group Top1/Top3 of 36.26%/67.03% and exact-region Top1/Top3 of 24.18%/42.86%. Hybrid exact Top3 exceeded logCPM baseline (30.77%) and projected-VSD-only scoring (29.67%). In TCGA/BraTS, hybrid Network Top3 was 40.00% and broad-anatomy Top3 was 64.62%, supporting only coarse anatomical consistency. GSE189919 overlapped 15,622/21,668 projector genes, corresponding to 72.10% gene-space coverage, supporting projection feasibility rather than source-localization accuracy.

Supplementary Figure S1. Main figure source artwork

cfRNA-BrainTrace three-tier Application Note summary



Supplementary Table 1. TableS1_internal_validation_design

Validation	Data	Held-out unit	Route tested	Reported endpoints
Network LOSO	Bo2023 macaque brain RNA-seq	Single sample	Projected VSD vs logCPM/native VSD Network scoring	Network Top1, Network Top3, median true-rank
Network LOMO	Bo2023 macaque brain RNA-seq	One monkey	Projected VSD vs logCPM/native VSD Network scoring	Network Top1, Network Top3, median true-rank
Formal three-tier LOSO	Bo2023 macaque brain RNA-seq	Single sample	Projected-VSD Network beam plus logCPM group/exact rerank	Network, resolution-group and exact-region Top1/Top3
Formal three-tier LOMO	Bo2023 macaque brain RNA-seq	One monkey	Fold-local Network, group and exact components	Network, resolution-group and exact-region Top1/Top3

Supplementary Table 2. TableS2_internal_validation_results

Dataset	Route	Endpoint	Top1	Top3	Median true-rank	Interpretation
Bo2023 LOSO	Projected VSD Network	Network	58.00%	91.58%	1.0	Supports projected-VSD Network beam
Bo2023 LOMO	Projected VSD Network	Network	53.72%	91.33%	1.0	Supports donor-level Network beam
Bo2023 LOSO	Formal hybrid	Network	58.35%	92.38%	1.0	Primary endpoint
Bo2023 LOSO	Formal hybrid	Resolution group	44.47%	72.36%	2.0	Main region-level endpoint
Bo2023 LOSO	Formal hybrid	Exact region	22.48%	45.33%	4.0	Exploratory ranking
Bo2023 LOMO	Formal hybrid	Network	57.75%	91.21%	1.0	Cross-monkey support
Bo2023 LOMO	Formal hybrid	Resolution group	41.38%	69.09%	2.0	Main region-level endpoint
Bo2023 LOMO	Formal hybrid	Exact region	22.17%	42.36%	5.0	Exploratory ranking

Supplementary Table 3. TableS3_external_validation_design

Dataset	Sample type	n	Truth label type	Allowed conclusion
AHBA	Human normal brain RNA-seq	242 total; 233 supported; 91 exact-evaluable	Mapped anatomical labels	Cross-species mapped-label validation
TCGA/BraTS	Glioma tissue RNA-seq with MRI-derived labels	65 patients	Human imaging labels	Coarse anatomical consistency only
GSE189919	External count matrix	51 samples	No patient-level anatomical truth	Projection feasibility only

Supplementary Table 4. TableS4_external_validation_results

Dataset	Route	Endpoint	Top1	Top3	Conclusion
AHBA	Hybrid	Network	74.68%	94.42%	Strong mapped-label Network support
AHBA	Hybrid	Resolution group	36.26%	67.03%	Moderate group-level support
AHBA	Hybrid	Exact region	24.18%	42.86%	Only exact-evaluable mapped labels
AHBA	logCPM baseline	Exact region	17.58%	30.77%	Below hybrid exact Top3
AHBA	Projected VSD only	Exact region	10.99%	29.67%	Below hybrid exact Top3
TCGA/BraTS	Hybrid	Network	15.38%	40.00%	Coarse consistency only
TCGA/BraTS	Hybrid	Broad anatomy	13.85%	64.62%	Coarse consistency only
GSE189919	Projection feasibility	Projector gene overlap	15622/21668	72.10%	No accuracy claim

Supplementary Table 5. TableS5_figure_table_index

Item	Content	Use in manuscript
Figure 1	Formal route diagram and validation summary	Main Application Note figure
Supplementary Figure S1	Main figure source artwork reproduced in the supplementary PDF	Supplementary visual reference
Table S1	Internal validation design	Documents LOSO/LOMO operations
Table S2	Internal validation results	Reports Network, group and exact-region metrics
Table S3	External validation design	Documents label support and allowed conclusions
Table S4	External validation results	Reports AHBA, TCGA/BraTS and GSE189919 outcomes

Supplementary Table 6. TableS6_claim_boundaries

Avoid	Use
Projected VSD creates a new Bo2023 atlas.	Only query profiles are projected for Network beam generation.
Projected VSD is best for exact-region inference.	Projected VSD supports Network beam; logCPM supports downstream reranking.
TCGA/BraTS validates Bo2023 exact regions.	TCGA/BraTS supports coarse anatomical consistency.
GSE189919 validates accuracy.	GSE189919 verifies projection feasibility only.
Exact Top1 is deterministic localization.	Exact outputs are exploratory local candidate rankings.